

A buck converter is a switching regulator that converts a higher DC voltage into a lower DC voltage. Voltage dividers and linear regulators similarly reduce voltage, but operate differently. Voltage dividers establish reference voltages, but are unsuitable for delivering consistent power as changes to the load cause voltage changes. Linear regulators offer a stable, low-noise output, but their pass transistor continuously carries the load current supporting the difference between the input and output voltage, dissipating power $P_{loss} = (V_{in} - V_{out}) I_{out}$ as heat.

Alternatively, a buck converter efficiently transfers power via PWM, consisting of a controlled switch, a freewheeling diode, an inductor, an output capacitor, and a given load, which is the circuitry powered by the buck converter's stepped-down voltage.The switch is controlled using pulse-width modulation (PWM), meaning that it repeatedly alternates between ON and OFF states. The duty cycle denoted as D represents the fraction of each switching period T for which the switch remains on, such that $D = T_{ON} / T$, where T_{ON} is when the switch is connected to the circuit and in an "ON" state, and T is the switching period. This duty cycle is a ratio from $0<D<1$, and an increased duty cycle means the input voltage V_{IN} is connected to the inductor for a longer relative time, increasing average output voltage by the relationship $V_{OUT} = (D)(V_{IN})$. Thus, rather than dissipating the voltage difference as heat, an ideal buck converter conserves power by exchanging reduced voltage for increased output-current capability, seen by $P_{IN} = P_{OUT}$.

To understand voltage and current's behavior in a buck converter, the circuit must be examined during the switch's ON and OFF states. The voltage and current waveforms in a buck converter alternate between distinct ON and OFF behaviors because opening the switch changes both the current path and the source of energy powering the load. When the switch is ON, conventional current flows from the input source through the switch and inductor before reaching the output node, where KCL requires the inductor current to split according to $i_L = i_C + i_{LOAD}$. Applying KVL around the main ON-state loop gives $V_{IN} - V_L - V_{OUT} = 0V$; rearranging this equation gives $V_L = V_{IN} - V_{OUT}$, where V_L is the voltage drop across the inductor. Because V_{IN} is greater than V_{OUT}, V_L is positive. Inductors relate current and voltage via $V_L = L (dI_L / dt)$, so positive V_L means dI_L/dt is positive and current ramps upward. The source powers the load while storing energy in the inductor and when $i_L > i_{LOAD}$, i_C is positive, and the capacitor is charged. When the switch turns OFF, the source is disconnected, but the inductor maintains current in the same direction by reversing its voltage polarity. This forward-biases the diode, creating a path through the diode, inductor, and output. KVL now gives $V_L + V_{OUT} = 0$, so $V_L = -V_{OUT}$ ($V_C = V_{OUT}$ because connected in parallel). The inductor current ramps downwards, releasing stored energy. Still, $i_C = i_L - i_{LOAD}$, as the capacitor charges with excess current and The capacitor discharges when the inductor supplies insufficient current to the load. Thus, the diode maintains a current path, and the inductor converts the rectangular switching voltage into a triangular current ripple, which the capacitor smooths into a DC output voltage.

Switching frequency, $f_S = 1/T$, is the number of ON-and-OFF cycles completed per second. Higher frequencies shorten each interval, giving the inductor and capacitor less time to change current and voltage, reducing ripple and permitting smaller components. A lower frequency causes fewer switch transitions, reducing switching losses, heat, and electrical noise. Thus, frequency selection balances efficiency and noise against ripple, component size, and output stability.